

Carolin **Brandt**

Assistant Professor in Software Engineering

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Academic Employment & Education

Assistant Professor
September 2024 – current

Delft University of Technology, Software Engineering Research Group

- Considering the humans who will use our software and the developers who create it during software engineering processes
- Leading the research line on Human and Developer-Centered Software Engineering
- Instructor for the courses Software Quality and Testing (BSc), and Sustainable Software Engineering (MSc)

Lab Manager: FUSE Research Lab
2025 – current

Future of Software Engineering ICAI Lab: https://se.ewi.tudelft.nl/fuse-lab/

Board Member of DiversAbility
2023 – current

TU Delft Resource Group for Neurodivergence, Hidden and Chronic Vulnerabilities: https://tudelft.nl/diversability

Postdoctoral Researcher
February – August 2024

Delft University of Technology, Software Engineering Research Group

- Empowering automatic generation tools to collaborate with sotware developers
- Strengthening my experience with a focus on co-supervising PhD Students and international / industry research collaborations

Visiting Researcher
August – November 2023

Hamburg University, Applied Software Technology Group

- Guest of Prof. Walid Maalej, one of the world's experts in requirements engineering
- Exploring connections between software testing and requirements engineering research

PhD Student
2020 – 2024

Delft University of Technology, Software Engineering Research Group

- Graduated with distinction (cum laude, top 5% of graduates) in July 2024
- Test Amplification For and With Developers
- Part of the TestShift project, advised by Andy Zaidman and Arie van Deursen

Master

How to Analyze Build Logs — A Comparative Study of Chunk Retrieval Techniques
2019

Paper-Based Master Thesis, Guest at the Software Engineering Research Group, Delft University of Technology

- Composing a dataset of build logs labeled with extractable substrings
- Comparing Programming by Example, classical information retrieval and keyword search

Elite Graduate Program Software Engineering
2017 – 2019

University of Augsburg, Technical University of Munich, Ludwigs-Maximilans-University Munich — *Avg. grade: 1.09, 'very good'*

- Geared towards especially talented and motivated students in computer science
- Advanced education in software engineering, formal methods, database systems, distributed systems and human-computer interaction
- Highly selective admission: at most 20 students per year

Bachelor

A Description Language for Structural Smells
2017

Bachelor Thesis in Computer Science

- Developed a domain specific language to define automated analyses for structural smells within requirements artifacts
- Implemented automated analysis generation within the Qualicen Scout requirements analysis tool

Scholarship "Deutschlandstipendium"
2016 – 2018

Combined public and industrial scholarship for excellent students

Admission to best.in.tum
2015

Group of the best 2% of students at the faculty of computer science

Bachelor Computer Science: Games Engineering
2014 – 2017

Technical University of Munich — *Avg. grade: 1.3, 'very good'*

- Computer science with a practical focus in games
- Computer graphics, game physics, artificial intelligence, social games, human-computer interaction and usability

Publications

Journals

Shaken, Not Stirred. How Developers Like Their Amplified Tests — 2024

Brandt, Carolin; Khatami, Ali; Wessel, Mairieli; Zaidman, Andy

- IEEE Transactions on Software Engineering
- https://doi.ieeecomputersociety.org/10.1109/TSE.2024.3381015, preprint: https://carolin-brandt.de/publications/brandt-tse24.pdf

Developer-Centric Test Amplification: The Interplay Between Automatic Generation and Human Exploration — 2022

Brandt, Carolin; Zaidman, Andy

- Empirical Software Engineering 27.4 (2022): 1-35.
- https://doi.org/10.1007/s10664-021-10094-2 (open access)

The Qualitative Factor in Software Testing: A Systematic Mapping Study of Qualitative Methods — 2025

Baris Ardic · Carolin Brandt · Ali Khatami · Mark Swillus · Andy Zaidman

- Journal of Systems and Software
- preprint: https://carolin-brandt.de/publications/ardic-jss25.pdf

Conferences

End-to-End Regulatory Compliance Evolution Pipeline: From Legal Requirements to Developer Tests — 2026

Carolin Brandt; Sallam Abualhaija

- International Conference on Automated Software Engineering: New Ideas and Emerging Results (ASE'26)
- preprint: https://carolin-brandt.de/publications/brandt-ase26.pdf

Addressing Test Flakiness: Practical Approaches in a Database-Reliant Industrial System — 2026

George Vegelien · Carolin Brandt · Bas Graaf · Arie van Deursen

- ACM/IEEE International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP'26)
- Collaboration with Exact
- preprint: https://carolin-brandt.de/publications/vegelien-icseseip26.pdf

Beyond the YAML File: Understanding Real-World GitHub Actions Workflow Adoption — 2026

Ali Khatami, Carolin Brandt, Andy Zaidman

- International Conference on Evaluation and Assessment in Software Engineering (EASE'26)
- preprint: https://carolin-brandt.de/publications/khatami-ease26.pdf

Towards Refined Code Coverage: A New Predictive Problem in Software Testing — 2025

Carolin Brandt; Aurora Ramírez

- International Conference on Software Testing, Verification and Validation: Short Papers, Vision and Emerging Results (ICST'25)
- https://doi.org/10.1109/ICST62969.2025.10989028, preprint: https://carolin-brandt.de/publications/brandt-icst25.pdf

Not One to Rule Them All: Mining Meaningful Code Review Orders From GitHub — 2025

Abir Bouraffa · Carolin Brandt · Andy Zaidman · Walid Maalej

- International Conference on Evaluation and Assessment in Software Engineering (EASE'25)
- preprint: https://carolin-brandt.de/publications/bouraffa-ease25.pdf

Software Quality Assurance Analytics: Enabling Software Engineers to Reflect on QA Practices — 2024

Khatami, Ali; Brandt, Carolin; Zaidman, Andy

- IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM'24)
- https://doi.org/10.1109/SCAM63643.2024.00027, preprint: https://carolin-brandt.de/publications/khatami-scam24.pdf

Mind the Gap: What Working With Developers on Fuzz Tests Taught Us About Coverage Gaps — 2024

Brandt, Carolin; Castelluccio, Marco; Holler, Christian; Kratzer, Jason; Zaidman, Andy; Bacchelli, Alberto

- ACM/IEEE International Conference on Software Engineering: Software Engineering in Practice (ICSE-SEIP'24)
- Collaboration with Mozilla and the Unversity of Zurich
- https://doi.org/10.1145/3639477.3639721, preprint: https://carolin-brandt.de/publications/brandt-icseseip24.pdf

Using GitHub Copilot for Test Generation in Python: An Empirical Study — 2024

El Haji, Khalid; Brandt, Carolin; Zaidman, Andy

- ACM/IEEE International Conference on Automation of Software Test (AST'24)
- https://doi.org/10.1145/3644032.3644443, preprint: https://carolin-brandt.de/publications/elhaji-ast24.pdf

When to Let the Developer Guide: Trade-offs Between Open and Guided Test Amplification — 2023

Brandt, Carolin; Wang, Danyao; Zaidman, Andy

- IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM'23)
- https://doi.org/10.1109/SCAM59687.2023.00032, preprint: https://carolin-brandt.de/publications/guided_test_amplification_SCAM_2023.pdf

How Does This New Developer Test Fit In? A Visualization to Understand Amplified Test Cases *2022*

Brandt, Carolin; Zaidman, Andy

- IEEE Working Conference on Software Visualization (VISSOFT'22)
- https://doi.org/10.1109/VISSOFT55257.2022.00011, preprint: https://carolin-brandt.de/publications/TIG_paper_VISSOFT_2022.pdf

Fixing Continuous Integration Tests From Within the IDE With Contextual Information *2022*

Boone, Casper; Brandt, Carolin; Zaidman, Andy

- IEEE/ACM International Conference on Program Comprehension (ICPC'22)
- https://doi.org/10.1145/3524610.3527908, preprint: https://carolin-brandt.de/publications/TestAxis_icpc22.pdf

Naming Amplified Tests Based on Improved Coverage *2021*

Nijkamp, Nienke; Brandt, Carolin; Zaidman, Andy

- IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM'21)
- https://doi.org/10.1109/SCAM52516.2021.00036, preprint: https://carolin-brandt.de/publications/NATIC_paper.pdf

Removing Redundant Statements in Amplified Test Cases *2021*

Oosterbroek, Wessel; Brandt, Carolin; Zaidman, Andy

- IEEE International Working Conference on Source Code Analysis and Manipulation (SCAM'21)
- https://doi.org/10.1109/SCAM52516.2021.00037, preprint: https://carolin-brandt.de/publications/RS_paper.pdf

LogChunks: A Data Set for Build Log Analysis *2020*

Brandt, Carolin; Panichella, Annibale; Zaidman, Andy; Beller, Moritz

- International Conference on Mining Software Repositories (MSR'20).
- https://doi.org/10.1145/3379597.3387485 (open access)

Workshops

Keeping Humans in the Driver's Seat: A Conceptual Framework for GenAI and Competence Debt *2026*

Clara Marie Lüders · Olaf Krużycki · Carolin Brandt

- HumanAISE'26
- preprint: https://carolin-brandt.de/publications/lueders-humanaise26.pdf

On the Energy Cost of Static Analysis Precision: An Empirical Study of SpotBugs Effort Levels *2026*

Sophie van der Linden · Xutong Liu · Andy Zaidman · Carolin Brandt

- DevOpsSustain'26
- preprint: https://carolin-brandt.de/publications/vanderlinden-devopssustain26.pdf

The Energy Impact of Batch Testing in Continuous Integration: An Empirical Study of Static and Dynamic Batching Strategies *2026*

Máté Oszkó · Xutong Liu · Andy Zaidman · Carolin Brandt

- DevOpsSustain'26
- preprint: https://carolin-brandt.de/publications/oszko-devopssustain26.pdf

Strategies and Challenges in Recruiting Interview Participants for a Qualitative Evaluation *2022*

Brandt, Carolin; Zaidman, Andy

- International Workshop on Recruiting Participants for Empirical Software Engineering (RoPES'22)
- preprint: https://carolin-brandt.de/publications/ropes_2022.pdf

Incremental Just-In-Time Test Generation in Lock-Step with Code Development *2022*

Brandt, Carolin

- International Summer School on Search-and Machine Learning-based Software Engineering (SMILESENG)
- Received the Best "New Ideas and Work in Progress" Paper/Presentation Award
- preprint: https://carolin-brandt.de/publications/smileseng_2022_abstract.pdf

Pen-Pen: A Wellbeing Design to Help Commuters Rest and Relax *2018*

Pichlmair, Monika; Brandt, Carolin; Henrich, Marcel; Biederer, Alexander; Aslan, Ilhan; Bittner, Björn; André, Elisabeth

- Designed and evaluated a neck-cushion, mobile app and multi-modal feedback loop to increase commuters' wellbeing.
- ICMI Human-Habitat for Health Workshop, https://doi.org/10.1145/3279963.3279966

Prizes

2026 **Distinguished Reviewer Award**, International Conference on Software Maintenance and Evolution
2026 **Distinguished Reviewer Award**, International Conference on Cooperative and Human Aspects of Software Engineering
2025 **Distinguished Reviewer Award**, International Conference on Software Maintenance and Evolution
2025 **Distinguished Reviewer Award**, International Conference on Program Comprehension
2024 **Distinguished Reviewer Award**, International Conference on Mining Software Repositories
2023 **Distinguished Reviewer Award**, International Conference on Mining Software Repositories
2022 **Best "New Ideas and Work in Progress" Paper / Presentation**, SMILESENG Summer School
2022 **Award for Best Artifact**, IEEE Working Conference on Software Visualization
2022 **Second Place Poster Competition**, Alice & Eve 2022
2021 **International Women in Technology Scholarship**, Personal Scholarship by Zonta, US$8000
2017 **First Place at the HackaTUM Hackathon**, Category "Bike-Rental Systems Analysis"
2014 **Best Graduate in Chemistry**, German Chemical Society
2014 **Third Place at the Individual Competition of the "Day of Mathematics"**, University of Ulm

Supervised PhD Students

since 2026 **Mitrasee Deb**, Engineering Productivity Metrics
since 2025 **Mahdiyeh Barzegar**, Just-in-Time Test Generation
since 2024 **Mark Swillus**, Grounded Theory on Testing Strategies
2024-2026 **Ali Khatami**, Understanding Software Quality Assurance in Open-Source Communities: Awareness, Adoption, and Tool Utilization

Supervised MSc Students

current **Gopal Panchu**, on legacy modernization and advice in the public sector — *Master Thesis with the AcICT*
2026 **Mirko Boon**, on identifying test-worthy changes in pull requests — *Master Thesis*
2026 **Ilma Jaganjac**, on developer and AI interaction tracking in the IDE — *Master Thesis*
2026 **Roderick de Britto Heemskerk**, on predicting coverage in the Mozilla Firefox source code — *Master Thesis*
2025 **Nada Mouman**, on quality feedback for LLM-generated tests — *Master Thesis*
2025 **George Vegelien**, on reliably measuring flaky tests in industrial contexts — *Master Thesis with Exact*
2023 **Khalid El Haji**, on the usabiilty of generic LLM-generated tests — *Master Thesis*
2022 **Danyao Wang**, on user-directed test amplification — *Master Thesis*
2021 **Casper Boone**, on in-IDE feedback from continuous integration runs — *Master Thesis*

Academic Service

2026 **REFSQ**, Poster & Tools Co-Chair
2026 **VISSOFT**, Program Committee
2026 **SCAM**, PC Co-Chair
2026 **SANER Short Papers**, Program Committee
2024+25 **BENEVOL**, Program Committee
2025+26 **ICSME**, Program Committee
2026+27 **CHASE**, Program Committee
2025 **ICPC**, Program Committee
2025 **IDE Workshop**, Co-Organizer
2025 **ICST**, Publicity Chair
2024 **IEEE Software**, Journal Reviewer
2024 **ASE**, Artifact Evaluation Committee
2023+24 **MSR**, Junior Program Committee (2023), Program Committee (2024)
2023+24 **MSR: Data and Tool Showcase**, Program Committee
2024+25 **VST**, Program Committee
2023 **ACM Transactions on Software Engineering and Methodology**, Journal Reviewer
2022 **FUZZING**, Artifact Evaluation Committee
2022+23 **IEEE Transactions on Software Engineering**, Journal Reviewer
2022 **Journal of Systems and Software**, Journal Reviewer
2022 **BENEVOL, TestVis**, Program Committee
2022 **ECOOP**, Publicity Chair
2022 **Alice & Eve**, Web Chair
2021 **Journal of Software: Evolution and Process**, Journal Reviewer
2021 **VISSOFT**, Program Committee, Publicity Chair
2021 **ISSTA**, Artifact Evaluation Committee
2021 **ICSE**, Organization Committee, Watch Parties and Gather.Town
2020 **ICSE**, Student Volunteer, Streaming in the Europe Timeband
2020 **ESEC/FSE, ASE**, Student Volunteer

Practical Experiences

Research Industry Internship at Oracle Labs

2018

Two Months with the Multilingual Engine Team at Oracle Labs Zurich, Switzerland

Python, pandas and SQL

- Transformed pandas data analysis operations to SQL
- Benchmarked against python running within the database process using GraalVM

Software Engineer at Qualicen GmbH

2017 – 2018

Quality analyses of natural language requirements and tests artifacts

Java, ConQAT and Ranorex

- Extended ConQAT based analyses of requirements documents
- Halved the code size by safely deleting no longer used research components
- Developed automated UI test using Ranorex
- Managed GitLab and development workflow

Languages

fluent Dutch, English
native speaker German
elementary French, Japanese